Sujal Lohar

Full-Stack Software Engineer & DevSecOps Architect

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SUMMARY

Undergraduate researcher in Computer Science at Nirma University with peer-reviewed publications indexed in IEEE Xplore. Specializing in decentralized federated learning, differential privacy in clinical IoT, and explainable AI architectures. Selected delegate to Harvard HPAIR Tokyo 2025.

PEER-REVIEWED PUBLICATIONS

"Federated Transformer-Based Learning for Predicting ICU Events with Multi-hop XAI Explanation Models"

Sujal Lohar, Co-Authors, Dr. Aparna Kumari (Mentor)

IEEE Conference (Published in IEEE Xplore) • IEEE (2025) • [PUBLISHED]

"Privacy-Preserving Federated Healthcare Intelligence via Blockchain and Differential Privacy"

Sujal Lohar, Taher Gandhi, Yug Dave, Vineet Ameta, Dr. Aparna Kumari

COMSYS 2025: 6th International Conference on Frontiers in Computing and Systems • Springer / IEEE (Warsaw, Poland) (2025) • [ACCEPTED]

"Explainable Deep Learning Framework for Intensive Care Unit Anomaly Forecasting"

Sujal Lohar, Yash Raval, Utkarsh Mishra, Dr. Aparna Kumari

COMSYS 2025: 6th International Conference on Frontiers in Computing and Systems • Springer / IEEE (Warsaw, Poland) (2025) • [ACCEPTED]

"Multi-Hop Attention Rollout in Clinical Event Sequences"

Sujal Lohar, Taher Gandhi, Dr. Aparna Kumari

COMSYS 2025: 6th International Conference on Frontiers in Computing and Systems • Springer / IEEE (Warsaw, Poland) (2025) • [ACCEPTED]

"Intelligent Network Computing Paradigms for Distributed Clinical Data Systems"

Sujal Lohar, Co-Authors, Dr. Aparna Kumari

ICDSINC 2025: International Conference on Data Science and Intelligent Network Computing • NIT Raipur (2025) • [ACCEPTED]

EDUCATION

Institute of Technology, Nirma University

Aug 2023 – May 2027

*Bachelor of Technology (B.Tech) in Computer Science and Engineering (Minor in Cybersecurity) — **8.44 / 10.0 CGPA***

- Core Coursework: Distributed Systems, Operating Systems, Computer Networks, Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming (Java/C++), Cryptography & Network Security, Cloud Computing.
- Recipient: Ministry of External Affairs SPDC Full Scholarship (2023 - 2027).
- Competitive Programming: Codeforces Specialist (Rating 1400+), solve algorithmic graph and dynamic programming problems daily.

Swami Vivekanand Govt. Model School

Jul 2022 – May 2023

*Senior Secondary Certificate (Class XII) in Science Stream (PCM) — **95.8%***

- Awarded Rajasthan State Scholar Student Laptop Award for academic excellence in district rankings.

EXPERIENCE

Undergraduate Research Fellow | Institute of Technology, Nirma University

Aug 2024 – Present

Ahmedabad, Gujarat, India

- Engineered decentralized federated learning pipelines using FedProx across 40,000+ MIMIC-III ICU records.
- Attained 96.4% ICU hazard classification accuracy with multi-hop attention rollout for clinical interpretability.

- Co-authored 5 peer-reviewed papers indexed in IEEE Xplore and presented at international conferences.

Selected Delegate (HPAIR Tokyo) | Harvard Project for Asian and International Relations

Aug 2025 – Oct 2026

Tokyo, Japan

- Selected among international applicants (sub-10% acceptance rate) to represent India and Nirma University at Tokyo summit.
- Collaborated on technology policy and public infrastructure security frameworks.

SELECTED ENGINEERING PROJECTS

GovAsset Nexus: Polyglot Microservices Platform

Oct 2026

Stack: Java 21 • Spring Boot 3.2 • Redis 7.2 • gRPC • PostgreSQL • MongoDB • Kafka • Docker • Kubernetes • Istio

- Architected 5 polyglot microservices on Java 21 / Spring Boot 3.2 and Node.js using gRPC and Kafka.
- Engineered distributed Saga choreography with PostgreSQL transactional outbox for zero-loss recovery.
- Deployed to Kubernetes with Istio service mesh and Redis 7.2 sliding-window rate limiting.

Federated Clinical Predictor & XAI Engine

Aug 2025

Stack: Python 3.11 • PyTorch • FedAvg / FedProx • TabTransformer • MIMIC-III Dataset • Scikit-Learn • Flask • Docker

- Formulated decentralized federated learning pipeline using FedProx on 40,000+ MIMIC-III ICU records.
- Attained 96.4% predictive accuracy with multi-hop attention rollout for clinical feature interpretability.
- Co-authored research paper published in IEEE Xplore and presented at international conferences.

Edge IoT Health Telemetry Pipeline

May 2025

Stack: C++ • ESP32 • MQTT (TLS 1.3) • InfluxDB • Grafana • Docker • Python • WebSocket

- Programmed ESP32 C++ firmware for 100Hz physiological biometric sensor sampling.
- Engineered TLS 1.3 MQTT telemetry ingestion pipeline into InfluxDB and Grafana dashboards.

NASA Near-Earth Object Orbital Risk Engine

Oct 2024

Stack: Three.js • WebGL • TypeScript • Python FastAPI • NASA JPL API • Docker

- Constructed Three.js 3D orbital simulator processing 30,000+ near-Earth asteroids from NASA JPL APIs.
- Integrated Python hazard classifier predicting close-approach risk factors.

TECHNICAL SKILLS

Programming Languages: C++, Python, JavaScript / TypeScript, Java, SQL, Bash / Shell

Cloud, DevOps & DevSecOps: Docker, Kubernetes, GitHub Actions, DevSecOps (Trivy / GitLeaks), AWS & OCI, Linux System Administration

Backend & Distributed Systems: Spring Boot 3.2, Node.js & Express, NestJS, gRPC & Protocol Buffers, Kafka & Redis

Security, Cryptography & Observability: Cryptographic SHA-256 Ledger, Prometheus & Grafana, Falco Runtime Detection, Active Defense Honeypots

Machine Learning & Federated Systems: Federated Learning (FedAvg / FedProx), Explainable AI (XAI), MediaPipe Computer Vision, Scikit-Learn, NumPy & Pandas

CERTIFICATIONS & HONORS

Oracle Cloud Infrastructure 2024 DevOps Professional (Oracle Cloud Infrastructure, Oct 2024)

NASA Space Apps Challenge Global Nominee (NASA - National Aeronautics and Space Administration, Oct 2024)

Selected Delegate - HPAIR Harvard Conference (Harvard College Project for Asian and International Relations (Tokyo, Japan))

UG Research Fellowship Grant (10,000 INR) (UG Research Club, Nirma University)

Oracle Cloud Infrastructure 2024 Generative AI Certified Professional (Oracle Cloud Infrastructure, Aug 2024)

Top 8 Sanchar Mitras State Recognition (Department of Telecommunications (DoT), Government of India - Gujarat LSA)

Scholarship Programme for Diaspora Children (SPDC) (Ministry of External Affairs, Government of India)